

SANJAY J

13 Jamiya Nagar Kovaipudur , Coimbatore - 42

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Computer Science undergraduate specializing in AI & ML with strong skills in Java, Python, C/C++, and React. Experienced in research, mobile/web development, and leading technical events, with a passion for building efficient and scalable solutions.

COURSEWORK

- Data Structures • Object Oriented Programming • Design and Analysis of Algorithms • Operating Systems
- Database Management Systems • Computer Architecture • Software Engineering • Discrete Mathematics
- Probability and Statistics • Machine Learning • Computer Networks • Embedded Systems • Artificial Intelligence
- Hardware Programming • Optimisation Techniques

PROFESSIONAL EXPERIENCE

AIBILL Project – Inria Grenoble & PSG College of Technology

Jan 2025 - May 2025

Mobile Application Developer

- Contributed to front-end, back-end, and database development for a mobile app prototype. • Built cross-platform mobile app using React Native , Tailwind and Worked on backend APIs and MySQL database integration.
- Collaborated on system integration for an international startup project incubated at Inria.
- Contributed to the early-stage success of the AIBILL prototype in an international research industry

PROJECTS

- **Sudar AI** : AI-Powered Adaptive Learning Assistant , Developed an AI agent to generate personalized worksheets based on student performance, integrate teacher feedback, and adapt difficulty levels. Implemented visual storytelling for grades 1–5 using image generation and TTS, and created subject-specific animations for grades 6–12 with Manim. Enabled interactive doubt clearing, local-language AI conversations, and culturally sensitive teaching support, empowering teachers in rural and semi-urban India.
- **Command Gateway** : Built a full-stack Command Gateway system using React and FastAPI to securely manage command execution through regex-based rule enforcement, API-key authentication, and credit-based usage control. Implemented role-based access for admins and users, audit logging, and deployed the application on cloud infrastructure for real-world usage.
- **PSG - AI Consortium , AI Spectrum Conference and ICAMM 2026 Websites** : Built official websites for PSG - AI Consortium and AI Spectrum 2025, and ICAMM '26 (on process) using React and Tailwind , 3js. Implemented features like mobile responsiveness, CMS content integration, and SEO optimization.
- **Gender and Age Prediction** : Developed a deep learning model leveraging Convolutional Neural Networks (CNNs) to predict age and gender from facial images. Implemented image preprocessing and augmentation techniques to improve accuracy and robustness. Evaluated performance on benchmark datasets and optimized the model for real-world applicability.

RESEARCH

- **Neuro Prune : An addaptive Approach for efficient Deep Neural Network Optimisation on edge devices**
 - 1.Co-authored "Neuro Prune", a pruning algorithm optimizing DNNs for edge devices using reinforcement learning and the Lottery Ticket Hypothesis.
 - 2.Achieved 0.22% accuracy improvement on ResNet-18 with reduced inference time and computational load.
 - 3.Enabled energy-efficient, real-time AI deployment on resource-constrained environments.
 - 4.Presented this Research Paper at International Conference on Applied Mathematical Models 2025 titled " Neuro Prune : An addaptive Approach for efficient Deep Neural Network Optimisation on edge devices.

TECHNICAL SKILLS

- Programming Languages: Java , Python , C , C++
- Web Technologies: - HTML , CSS , Java Script , Typescript , React , Native , Tailwind , FastAPI
- Tools & Environments: Git, VS Code, Docker, Figma , Cursor, PyCharm , Netbeans

EDUCATION

PSG College of Technology

Aug 2023 – May 2027

B.E. Computer Science and Engineering (Artificial Intelligence & Machine Learning)

CGPA - 8.34* (up to 4th sem)

Coimbatore, Tamil Nadu

CS Academy Coimbatore CBSE

May 2021 – Apr 2023

Higher Secondary - 86.2% (Grade 12)

Coimbatore, Tamil Nadu

Nirmala Matha Convent School ICSE /ISC

Jun 2012 – Apr 2021

Secondary - 92% (Grade 10)

Coimbatore, Tamil Nadu

LEADERSHIP POSITIONS

- **Joint Secretary - Computer Science and Engineering Association** **Aug 2025 - present**
- **Head of Events - Cyber Security Club** **Sep 2025 - present**
- **Joint Secretary - Computer Science and Engineering Association** **Aug 2024 - May 2025**
 - Organising Secretary of Infinitum 2025 (Intra College Tech Symposium) featuring coding contests, tech talks, and project showcases.
 - Coordinator for 3 technical events, 3 workshops of CSEA, and 2 ceremonies.
 - Managed the yearly budget of CSEA and two affiliated clubs (GHCC and The Eye) for the academic year 2024–2025.
 - Coordinated 5 events conducted by CSEA for Kriya '25 (Students' Union Fest).